AP BIOLOGY · UNIT 1 · CHEMISTRY OF LIFE

Hydrogen Bond Bench

Two attractions pulling against each other in a glass tube, and what happens to energy that never becomes motion.

Name ____________ Date ____________ Period ____________

Demonstration: andresdbp.github.io/resources/demos/hydrogen-bond-bench/

1 Before you open anything

Answer from what you already know, then leave it alone. The point is to have a committed number on paper that the demonstration will either confirm or contradict.

PREDICTION

Water in a clean glass capillary of **0.80 mm** inside diameter climbs to some height *h*. The same water is now put in a clean glass capillary of **1.60 mm** inside diameter — exactly twice as wide, everything else identical. How does the new height compare with *h*? Give a ratio or a fraction, not “more” or “less”.

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In one sentence: what reasoning gave you that number?

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2 Guided setup

Open the demonstration. It has two tabs across the top of the stage: **1 · Capillary action** and **2 · Where energy goes**. Start on the first.

1. Set **Inside diameter** to **0.80 mm** and leave it there for the whole of this part.
2. Press the preset button **Water, clean glass**. Record the three readouts in the first row below, plus the small grey line underneath **Rise above the reservoir**.
3. Press **Water, waxed glass**, then **Mercury-like**, filling in the other two rows.

Preset	Cohesion (mJ m⁻²)	Adhesion (mJ m⁻²)	Surface tension γ	Contact angle θ	Rise above the reservoir	The grey line under the rise says…
Water, clean glass						
Water, waxed glass						
Mercury-like						

a. The first two rows have the *same* cohesion. Name the one quantity that differs between them, and say what it did to the sign of the rise.

b. Compare the surface tension γ you recorded with the cohesion in the same row. Write the relationship between them as an equation.

> Read the **Honest limits** panel before you write anything about mercury. The bench changes only the two bond strengths and holds the density at water's value the whole time, so the "Mercury-like" preset reproduces mercury's contact angle but not mercury's liquid. Real mercury sinks about 14 mm in a 0.80 mm tube, not the depth drawn here.

Now switch to the **2 · Where energy goes** tab. Two boxes of identical particles receive identical energy; only the right-hand box has hydrogen bonds available.

1. Set **Energy added per particle** to **1.00 arb. u.**
2. Set **Hydrogen bonds per particle** to **0** and fill in the first row.
3. Set it to **2.0** (roughly liquid water), then **4.0** (roughly ice), filling in the other two rows.

Bonds per particle	Motion energy, bonding off	Motion energy, bonding on	Fraction of the input that went into breaking bonds	Particle speed, on vs off
0				
2.0				
4.0				

c. The first row is the control. Say in one sentence what it establishes, and why the rest of the table would mean nothing without it.

d. Read the panel headed **Read this before you convert anything**. Write down the one thing it forbids you to do with these numbers, and the reason it gives.

3 The core relationship: bore and height

Back to the **1 · Capillary action** tab. Press **Water, clean glass** and do not touch either attraction slider again in this part — the liquid is now fixed and only the plumbing changes.

1. Move **Inside diameter** to each value in the table.
2. Record **Rise above the reservoir** each time.
3. Fill the last column by multiplying your two numbers together.

Inside diameter d (mm)	Rise above the reservoir h (mm)	$h \times d$
0.80		
1.00		
1.60		
2.00		
3.20		
4.00		

a. The last column does something the middle column does not. State the rule connecting h and d in your own words, in one sentence, and then as a proportionality.

b. Go back to your prediction in part 1. Was the ratio right? If not, name specifically the assumption that turned out not to hold.

c. The upward pull acts around the *circumference* of the liquid's contact with the glass; the weight it has to lift is a *volume*. Write how each of those scales with the radius r, and use them to explain the rule you just wrote.

> The lower graph, **How high it climbs against tube diameter**, is drawing the same rule as a curve. Check that the point it marks agrees with the row you just filled in before you move on.

4 Where the obvious move stops working

Adhesion is the attraction that touches the glass, so more of it should mean a taller column. That holds — up to a point. Find the point.

1. Set **Inside diameter** back to **0.80 mm**.
2. Set **Cohesion** to **144** and drag **Adhesion** up to its maximum, **160**. Record θ and *h*.
3. Leave adhesion at 160. Now drag **Cohesion** down through the values in the table, recording θ and *h* at each one.

Cohesion (mJ m^{-2})	Adhesion (mJ m^{-2})	Surface tension γ	Contact angle θ	Rise above the reservoir *h* (mm)
144	160			
120	160			
100	160			
60	160			

a. The contact angle column does not change. What value is it stuck at, and what physical situation does that value describe? (The note under the readouts names it.)

b. Adhesion never changed across those four rows, yet *h* fell steadily. So once θ has bottomed out, which attraction is holding the weight of the column up? Explain what each of the two attractions is doing at that point.

c. Water reaches the top of a hundred-metre tree through xylem. Using your answer to **b**, explain why the cohesion–tension mechanism needs *both* attractions and would fail with only one.

Now work backwards.

d. With cohesion at **144**, predict on paper the adhesion value that would make the liquid sit exactly level with the reservoir ($h = 0$). Write the prediction, then set the sliders and check it.

What does θ read at that setting, and what shape is the meniscus?

5 Solve these without the demonstration

Close the page, or look away from it. Show your working. Then reopen it, set the sliders and check yourself — and if a number is off, write down where. Use $\cos\theta = 2W_{adh}/W_{coh} - 1$, $\gamma = W_{coh}/2$, and $h = 2\gamma\cos\theta/(\rho g r)$ with $\rho = 997$ kg m^{-3} and $g = 9.81$ m s^{-2}.

1. A liquid has a work of cohesion of 120 mJ m^{-2} and a work of adhesion to glass of 90 mJ m^{-2}. It stands in a tube of 1.00 mm inside diameter. Find γ, $\cos\theta$, θ, and the rise *h* in millimetres.

2. A different liquid has a work of cohesion of 130 mJ m^{-2}. In a glass tube it sits exactly level with the reservoir, at any bore. Find its work of adhesion to glass and its contact angle, and state what shape the meniscus has.

3. Water on perfectly clean glass rises 36.8 mm in a tube of 0.80 mm inside diameter. Predict the rise in a tube of 3.20 mm inside diameter, to two significant figures, *without* recomputing from γ.

4. Two boxes of identical particles are each given 1.00 arbitrary unit of energy per particle. The left box has no hydrogen bonds; the right box averages two per particle. Explain (a) why a thermometer placed in the right-hand box would read lower, (b) why the missing energy is not lost, and (c) why the ratio between the two boxes must not be quoted as water's specific heat capacity in J g^{-1} K^{-1}. No arithmetic is required.

6 On your own

Nothing here is graded, and the demonstration does considerably more than this worksheet used.

- On the energy tab, hold **Hydrogen bonds per particle** at 2.0 and push **Energy added per particle** all the way to 3.50. Watch **Bonds broken** and the fraction going into bonds. One of them keeps climbing and one of them stops. Work out why, then predict what happens to the gap between the two boxes if you kept adding energy past the end of the slider.
- Drag the energy slider to exactly 0 and read the split bars. Then argue from the equation **$e = T + (n/2)\,\varepsilon\, x(T)$** why zero in has to mean zero everywhere, with nothing stored.
- On the capillary tab, the lower graph carries a dashed reference line for water on perfectly clean glass. Find a cohesion and adhesion pair that puts your curve exactly on top of it, and a pair that puts it exactly on the zero line across the whole width.
- The tube drawing announces how many times wider the bore is drawn than the height scale beside it. Change the diameter and watch that number. Why does the drawing have to lie about the bore, and what would a truthful drawing of a 0.80 mm tube look like?